

Conclusion

Summary of Contributions

This paper has applied the Cognitive Integrity Framework (CIF) [?] across ten critical domains, demonstrating that Goal Hijacking is not a narrow linguistic exploit but a structural corruption of autonomous decision-making. The specific contributions are:

C1: CIF-AD-OODA Integration Model. We formalized the integration of three complementary frameworks—CIF (defense mechanisms), Axiomatic Design (structural analysis) [?], and the OODA Loop (temporal dynamics) [?—into a unified analytical model for Goal Hijacking. This model enables systematic domain analysis through a standardized five-step template.

C2: Universal Attack Pattern Taxonomy. Through cross-domain analysis, we identified three universal attack patterns—FR Polarity Inversion, Constraint Relaxation, and Context Boundary Violation—that characterize all Goal Hijacking attacks as specific manipulations of the Axiomatic Design Matrix. FR Polarity Inversion is the most prevalent (5/10 domains), revealing that the most effective attacks *co-opt* rather than *disable*

Relationship to the Series

This paper completes a four-part arc in the Cognitive Security series:

- ▶ **Paper 1** (Theory) [?] established the formal foundations: cognitive state model, trust calculus, adversary taxonomy (Ω_1 – Ω_5), and five canonical defense mechanisms with composition algebra.
- ▶ **Paper 2** (Computation) [?] provided computational validation: benchmark evaluation across 950 attack scenarios, demonstrating 99.5% detection with the recommended defense stack.
- ▶ **Paper 3** (Practice) [?] grounded the framework in biological analogy: eusocial insect colonies as existence proofs for CIF-like defense mechanisms evolved over millions of years.
- ▶ **Paper 4** (Applications—this paper) demonstrates real-world applicability: CIF's formal mechanisms map to concrete defenses across ten high-stakes domains, and the cross-domain analysis yields new insights (universal attack patterns, novel defense extensions) not visible from any single domain

Future Work

Several directions emerge from this analysis:

1. **Empirical validation.** The most critical next step is controlled experimentation in at least one domain—ideally cyber-security or infrastructure, where testbed environments exist—to validate CIF defense effectiveness against real Goal Hijacking attacks with measured detection rates and false positive costs. The real-world incidents cataloged in Supplementary Material S2 provide natural experiment data for retrospective validation—particularly the Replit and procurement agent cases, where the full attack chain is documented and the hypothesized CIF defenses can be simulated against the recorded agent behavior.
2. **Multi-domain attacks.** Adversaries operating across domain boundaries (e.g., manipulating food security data to influence trade policy agents) represent a class of attacks that single-domain analysis cannot capture. Federated CIF architectures with cross-domain trust management are needed.

Closing Statement

Prior to this work, the threat of Goal Hijacking was largely viewed as an issue of content moderation—preventing a chatbot from saying something inappropriate. We have demonstrated that in the domain of deployed multiagent systems, Goal Hijacking is a kinetic and existential threat. It is the ability of an adversary to rewrite the Functional Requirements of our critical infrastructure, turning our own autonomous agents against us.

By integrating Axiomatic Design principles with the Cognitive Integrity Framework and analyzing the temporal dynamics through the OODA lens, we establish both a theoretical foundation and a practical defense methodology. We move from fragile “prompt engineering” to robust “goal engineering.” We secure the OODA loop not by sanitizing the world of adversarial inputs, but by hardening the agent’s Orientation phase against the transient seduction of the hijack—through Cognitive Firewalls that filter, Belief Sandboxes that isolate, Behavioral Invariants that constrain, Drift Detectors that monitor, and Byzantine Consensus that validates. In doing so, we ensure that as our systems become faster